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Survey of Radiation Exposure from Portable X-Ray Fluorescence

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INTRODUCTION

Portable X-ray Fluorescence (pXRF) analysis is an analytical technique that allows for the almost instant determination of the chemical composition of a wide range of samples. The technique is non-destructive, and the equipment is lightweight and easily portable, making it highly useful in various situations. Recent studies suggest that the application of pXRF for multi-elemental analysis in blood samples (dripped onto paper) shows very promising for clinical practice. Despite the low levels of radiation emitted by these devices, caution must be taken in their handling. The objective of this study is to conduct a radiometric survey of an experimental setup for pXRF analysis of blood samples.

MATERIALS AND METHODS

The X-ray Fluorescence analysis was performed using a compact X-ray spectrometer model X-123 SDD with Ag target. The experimental setup consisted of an compact X-ray spectrometer (Amptek) with Ag target, a semiconductor detector X-123 SDD, and an acrylic stand, where Whatman N.42 paper were inserted. Measurements were performed using a calibrated Radcal® ionization chamber, model 10x5, the 1800 cm^3 volume, which has application for radiation protection.

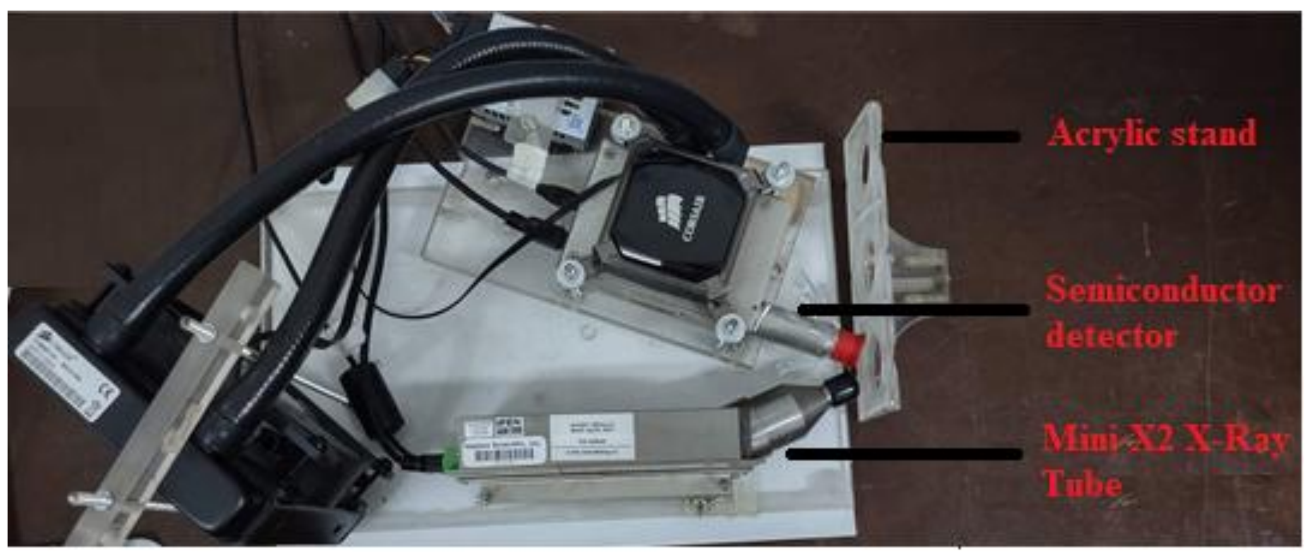

Figure 1 - Mini- X-Ray Tube and SDD detector

Figure 2 - Setup for radiometric survey of the pXRF spectrometer

The measurements were performed under two conditions: for the maximum current and voltage values of the equipment, and for the current and voltage values of the blood sample analysis protocol. Measurements were performed with the ionization chamber at various distances, 21, 35, 100 and 200 cm, and varying positioning in relation to the source.

RESULTS

The radiometric survey of the X-ray spectrometer, using the Mini-X2 tube, allowed fundamental conclusions to be drawn for the safe operation of the equipment. Analysis along the primary beam axis robustly validated the Inverse Square Law of Distance as the predominant factor for dose attenuation. Increasing the distance from 21 cm to 100 cm resulted in a dose rate reduction to 5.8% of the original value, a figure that is in excellent agreement with the theoretically calculated 4.4%. This result confirms that distance is the most effective radiation protection measure against the primary beam.

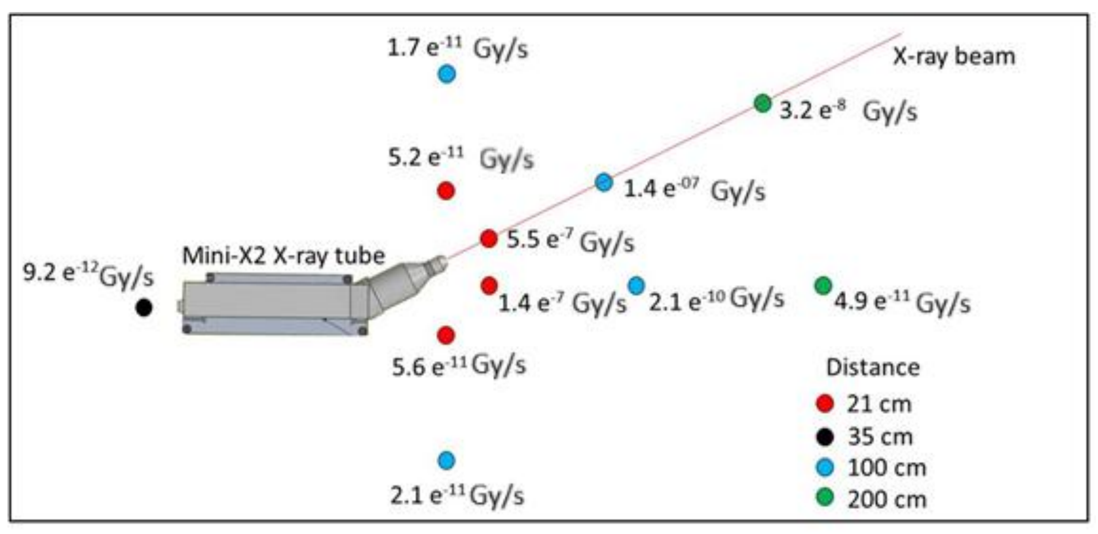

Figure 3 - Radiometric survey of the X-ray spectrometer

Position	Air Kerma Rate (Gy/s) for each Detector Source Distance				Uncertainties (%)
	21 cm	35 cm	100 cm	200 cm	
Direct beam	5.5×10^{-7}	-----	1.4×10^{-7}	3.2×10^{-8}	3,0
Behind pXRF	-----	9.2×10^{-12}	-----	-----	3,0
Front of pXrf	1.4×10^{-7}	-----	2.1×10^{-10}	4.9×10^{-11}	3,0
Next to pXRF (direct beam)	5.2×10^{-11}	-----	1.7×10^{-11}	-----	3,0
Next to pXRF (opposite to direct beam)	5.6×10^{-11}	-----	2.1×10^{-11}	-----	3,0

Table 1 - Radiometric survey of the X-ray spectrometer

These results confirm the equipment's operational safety and compliance with optimization principles for the tested conditions. However, to ensure full reliability, additional testing is needed to cover all possible operating conditions. Future studies should analyze radiation distribution across different geometries and assess accumulated dose in prolonged use to fully characterize the equipment's safety.

CONCLUSIONS

Preliminary measurements demonstrated the inherent radiation safety of the X-ray fluorescence spectrometry (pXRF) system under investigation. The measurements validated two fundamental protection principles: the effectiveness of the distance, with a clear attenuation of the primary beam dose following the Inverse Square Law (a reduction of approximately 6% of the value at 21 cm when moving away to 100 cm), and the effectiveness of the head shielding.

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